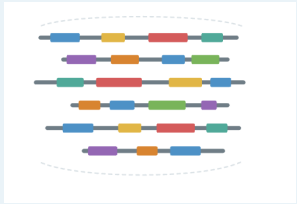
A ORF calling, clustering, and abundance-weighted prioritization

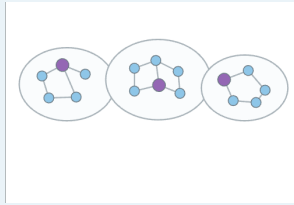
1



Raw insert sequences

NTC-selected inserts from PacBio sequencing

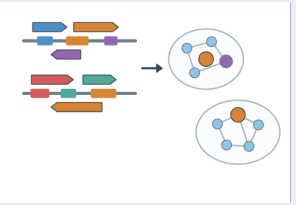
2



Insert clustering → RepSeqᵢ

MMseqs2: 99% identity and 99% coverage; retain cluster size n_i

3



ORF calling and clustering

Prodigal meta; ORFs clustered at 99% identity and 80% coverage

4

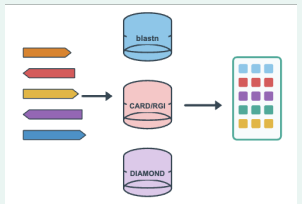


Abundance-weighted rank

RepORF support $W_k = \sum_{c \in C_k} n_c$; top 5% per library

B Function prediction, family placement, and validation

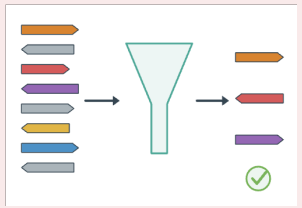
1



Parallel annotation

blastn→nt; CARD/RGI; DIAMOND→Swiss-Prot

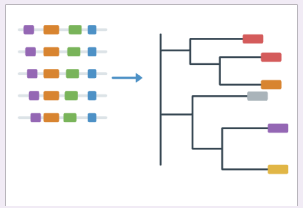
2



Evidence-based filtering

Integrate support, completeness, identity, coverage, and family relevance

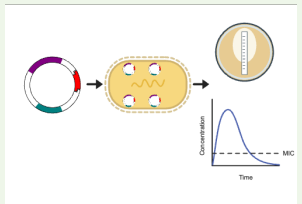
3



Family placement

MAFFT → trimAl → IQ-TREE 3 ModelFinder → ggtree

4



Causal validation

Clone prioritized ORFs; test MIC / IC_{50} against controls